

# PowerLine Command - Demo Version v1.5

## IMPLEMENTATION DOCUMENTATION

### OVERVIEW

This document explains how **Demo Version v1.5** was created from **Demo Version v1.4**, following the strict requirements in **Laws v1.5** and **Overview v1.5**.

### COMPLIANCE WITH LAWS v1.5

#### Version Alignment

- **Laws:** v1.5 ✓
- **Overview:** v1.5 ✓
- **Demo:** v1.5 ✓

All three documents are synchronized and follow the same version number.

#### Key Laws Followed

- LAW 1:** ✓ Working from existing Demo v1.4 (Blank Skin equivalent for dataset swap)
- LAW 2:** ✓ No core game logic modified
- LAW 3:** ✓ Dataset applied exactly as provided in CSV
- LAW 4:** ✓ All objects and mechanics preserved
- LAW 5:** ✓ Dataset integrity maintained - all challenges match specifications
- LAW 7:** ✓ Holodeck remains disabled (Demo Mode)
- LAW 8:** ✓ This is a Demo Skin (not Publish Mode)

### CHANGES FROM v1.4 TO v1.5

#### 1. Version Number Updates

| Location       | v1.4                         | v1.5                           |
|----------------|------------------------------|--------------------------------|
| HTML Title     | <title>PLC - Demo</title>    | <title>PLC - Demo v1.5</title> |
| CSS Header     | POWERLINE BLANK BASE VERSION | POWERLINE BLANK BASE VERSION   |
| Comment        | v1.4                         | v1.5                           |
| Laws Reference | (per Laws v1.4)              | (per Laws v1.5)                |

#### 2. Dataset Replacement

##### v1.4 Dataset (Blank):

```
const gameData = [
  { challenge: "", command: "", primaryTarget: "", additionalDetail: "" }
];
```

v1.5 Dataset (Roman Numerals 1-10):

```
const gameData = [
  { challenge: "I", command: "1", primaryTarget: "", additionalDetail: "" },
  { challenge: "II", command: "2", primaryTarget: "", additionalDetail: "" },
  { challenge: "III", command: "3", primaryTarget: "", additionalDetail: "" },
  { challenge: "IV", command: "4", primaryTarget: "", additionalDetail: "" },
  { challenge: "V", command: "5", primaryTarget: "", additionalDetail: "" },
  { challenge: "VI", command: "6", primaryTarget: "", additionalDetail: "" },
  { challenge: "VII", command: "7", primaryTarget: "", additionalDetail: "" },
  { challenge: "VIII", command: "8", primaryTarget: "", additionalDetail: "" },
  { challenge: "IX", command: "9", primaryTarget: "", additionalDetail: "" },
  { challenge: "X", command: "10", primaryTarget: "", additionalDetail: "" }
];
```

3. What Was NOT Changed

- ✓ All CSS styling (colors, layouts, animations)
- ✓ All game mechanics (spawn, collision, scoring)
- ✓ All object definitions (Target, Tower, Zone, etc.)
- ✓ All projectile logic
- ✓ All impact logic
- ✓ All difficulty scaling
- ✓ All screen management
- ✓ All input handling
- ✓ Holodeck code (remains disabled)

 DATASET STRUCTURE

Source File

**File:** Roman\_Numerals\_1\_-\_10.csv  
**Format:** CSV with headers (Challenge, Command)  
**Entries:** 10 challenge/command pairs

Dataset Properties

| Property           | Value               | Notes                                        |
|--------------------|---------------------|----------------------------------------------|
| Challenge Set Name | Roman Numerals 1-10 | Numbers 1 through 10 in Roman numeral format |
| Total Entries      | 10                  | I, II, III, IV, V, VI, VII, VIII, IX, X      |

| Property          | Value       | Notes                                           |
|-------------------|-------------|-------------------------------------------------|
| Primary Target    | (empty)     | Not used in this dataset                        |
| Additional Detail | (empty)     | Not used in this dataset                        |
| Difficulty Level  | Beginner    | Basic number recognition                        |
| Purpose           | Educational | Learn Roman numeral to Arabic number conversion |

Challenge Mapping

| Roman Numeral | Arabic Number | Difficulty                  |
|---------------|---------------|-----------------------------|
| I             | 1             | Simple                      |
| II            | 2             | Simple                      |
| III           | 3             | Simple                      |
| IV            | 4             | Moderate (subtraction rule) |
| V             | 5             | Simple                      |
| VI            | 6             | Simple                      |
| VII           | 7             | Simple                      |
| VIII          | 8             | Simple                      |
| IX            | 9             | Moderate (subtraction rule) |
| X             | 10            | Simple                      |

 GAMEPLAY WITH ROMAN NUMERALS

How It Works

1. **Target Spawns** - Green rectangle appears at top of screen
2. **Challenge Displays** - Yellow rectangle shows Roman numeral (e.g., "VII")
3. **Player Input** - Player must type the Arabic number in blue input box (e.g., "7")
4. **Correct Answer** - Tower fires projectile at Target, explosion occurs, points awarded
5. **Incorrect Answer** - Target fires back, shield lost, streak resets
6. **Missed Target** - Target reaches Zone, Impact Zone occurs, shield lost

Educational Value

This dataset teaches:

- **Roman Numeral Recognition** - Visual identification of Roman numerals
- **Number Conversion** - Translation between numeral systems
- **Pattern Recognition** - Understanding additive (VI = 6) vs subtractive (IV = 4) rules
- **Speed & Accuracy** - Building muscle memory under time pressure

Difficulty Progression

As the game progresses:

- **Speed Increases** - Targets fall faster (every 1 correct = 7% faster)
- **More Targets** - Additional simultaneous targets (every 10 correct = +1 target)
- **Mental Pressure** - Must recall conversions quickly
- **Pattern Testing** - IV and IX test subtraction rule understanding

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 TECHNICAL IMPLEMENTATION

Dataset Integration

The CSV data was converted to JavaScript array format:

- Each row became an object with `challenge` and `command` properties
- `primaryTarget` and `additionalDetail` fields left empty (reserved for future use)
- Array contains 10 entries matching the CSV exactly

Game Logic Compatibility

The Roman Numerals dataset works with existing game logic because:

- **Validation** - Command input compared to `data.command` field
- **Display** - Challenge text shown in yellow rectangle from `data.challenge` field
- **Randomization** - Game selects random entries from 10-item array
- **Uniqueness** - Prevents duplicate challenges on screen simultaneously

No Code Changes Required

Because the game was designed with dataset flexibility:

- Data structure matches template (challenge, command, primaryTarget, additionalDetail)
- No hardcoded values in game logic
- Spawn system works with any dataset size
- Input validation is string-based (works with numbers as strings)

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 DIFFICULTY SCALING WITH THIS DATASET

Scaling Behavior

| Metric      | Initial    | After 10 Correct | After 25 Correct |
|-------------|------------|------------------|------------------|
| Fall Speed  | 15 seconds | ~7 seconds       | ~3.5 seconds     |
| Spawn Rate  | 1500ms     | ~900ms           | ~500ms           |
| Max Targets | 3          | 4                | 5+               |
| Mental Load | Low        | Medium           | High             |

### Challenge Intensity

With only 10 unique challenges:

- **Repetition** - Players will see same numerals multiple times
- **Memorization** - Encourages learning through repetition
- **Speed Focus** - Less about "what is the answer" and more about "how fast can I type it"
- **Muscle Memory** - Builds automatic response to Roman numerals

## 🎮 VISUAL PRESENTATION

Default Colors (Unchanged from v1.4)

| Object        | Color            | Purpose                      |
|---------------|------------------|------------------------------|
| Targets       | Green (#00ff00)  | Falling objects to clear     |
| Challenges    | Yellow (#ffff00) | Display Roman numerals       |
| Command Input | Blue (#0000ff)   | Where player types numbers   |
| Zone          | Brown (#8b4513)  | Protective barrier at bottom |
| Tower         | Gray (#808080)   | Player's defense element     |

### Challenge Display Example

VII

← Yellow rectangle with Roman numeral

↓

[Target]

← Green rectangle (falling)

↓

↓ (falls toward Zone)

↓

—————

← Brown Zone line

Player types: 7 (in blue input box)  
Tower shoots projectile at Target  
Explosion occurs at collision point

## 🔒 DEMO MODE STATUS

### Mode Verification

- **Current Mode:** Demo Mode ✓
- **Holodeck Status:** Disabled ✓

- **Skin Type:** Demo Skin ✓
- **Publish Mode:** NOT ENGAGED ✓

Holodeck Activation Rule (LAW 7)

Per Laws v1.5, Holodeck (back door feature) will only be enabled when:

1. Publish Mode is engaged
2. Developer explicitly instructs activation

**Status:** Holodeck code present but inactive in this Demo version.

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 DATASET METADATA

CSV Source Information

**File Name:** Roman\_Numerals\_1\_-\_10.csv  
**Character Encoding:** UTF-8 (with BOM)  
**Delimiter:** Comma  
**Headers:** Challenge, Command  
**Data Rows:** 10  
**Empty Fields:** primaryTarget, additionalDetail (reserved for future use)

Data Validation

All entries validated for:

- ✓ Proper Roman numeral syntax (I, V, X, and subtraction rules)
- ✓ Correct Arabic number equivalents (1-10)
- ✓ No duplicates
- ✓ Sequential ordering (1 through 10)
- ✓ Standard Roman numeral conventions

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 TESTING RECOMMENDATIONS

Test Cases

1. **Basic Conversion Test**
  - Verify each Roman numeral displays correctly
  - Confirm correct Arabic number input clears target
  - Check incorrect input triggers Target Impact
2. **Edge Cases**
  - Test IV (4) - subtraction rule
  - Test IX (9) - subtraction rule
  - Test rapid sequential inputs
  - Test backspace behavior (should mark as imperfect)

3. **Game Mechanics**

- Verify scoring with this dataset
- Check difficulty scaling
- Test spawn randomization
- Confirm collision detection

4. **Visual Display**

- Ensure Roman numerals are readable in yellow rectangles
- Check challenge positioning
- Verify proper alignment with targets

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 **NEXT STEPS**

Immediate Use

This v1.5 Demo is ready for:

- ✓ Testing Roman numeral dataset
- ✓ Educational gameplay
- ✓ Difficulty balancing feedback
- ✓ User experience testing

Future Enhancements

Possible expansions:

1. **Extended Dataset** - Roman numerals 11-100
2. **Mixed Difficulty** - Combine simple and complex numerals
3. **Reverse Mode** - Show Arabic number, type Roman numeral
4. **Themed Skin** - Apply visual theme while keeping dataset
5. **Multiple Datasets** - Allow player to select which set to practice

Dataset Library

This demonstrates the template for future datasets:

- PowerLine commands (actual CAD functions)
- Math facts (multiplication tables)
- Vocabulary (word definitions)
- Code syntax (programming commands)
- Any challenge/response pair

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 **REFERENCE**

Version Synchronization

| Document | Version | Status |
|----------|---------|--------|
|----------|---------|--------|

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| Document     | Version | Status    |
|--------------|---------|-----------|
| Laws         | v1.5    | ✓ Current |
| Overview     | v1.5    | ✓ Current |
| Demo HTML    | v1.5    | ✓ Current |
| Build Report | v1.5    | ✓ Current |

File Locations

- **Source Dataset:** Roman\_Numerals\_1\_-\_10.csv
- **Demo Game:** Demo\_\_PLC\_\_v1\_5.html
- **Build Report:** 3\_\_Build\_Report\_\_PLD\_\_v1\_5.md
- **Laws:** 1\_\_Laws\_v1\_5.docx
- **Overview:** 2\_\_Overview\_\_PCL\_\_v1\_5.docx

☒ FINAL CHECKLIST

- ☒ Version numbers updated (v1.4 → v1.5)
- ☒ Dataset replaced (Blank → Roman Numerals)
- ☒ Laws v1.5 compliance verified
- ☒ Core game logic unchanged
- ☒ All objects preserved
- ☒ Visual styling unchanged
- ☒ Holodeck remains disabled (Demo Mode)
- ☒ Dataset integrity maintained
- ☒ Documentation complete
- ☒ Ready for testing

**Document Created:** 2025-01-10  
**Author:** Claude (Anthropic)  
**Purpose:** Build Report for Demo v1.5 with Roman Numerals Dataset  
**Compliance:** Laws v1.5, Overview v1.5